IGCSE PHYSICS 0625 · TOPIC 2

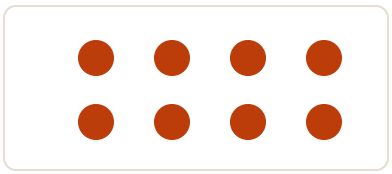
# Thermal Physics

The kinetic particle model, thermal properties, and how heat moves. Key definitions, formulas and graphs — distilled from the syllabus and past papers.

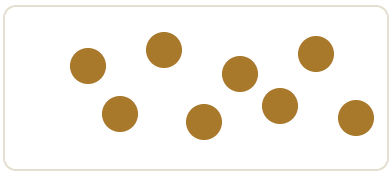
# Kinetic Particle Model of Matter

2.1

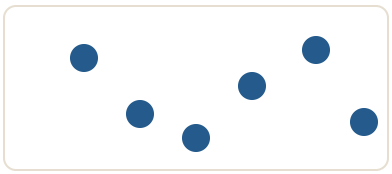
## STATES OF MATTER



**Solid** — particles tightly packed in a regular lattice, vibrate about fixed positions.



**Liquid** — particles close but disordered, slide past one another.



**Gas** — particles far apart, moving fast and randomly.

## DEFINITIONS

**Internal energy** the kinetic energy and potential energy of the molecules (particles) of the object

📌 考过 · ■ 42/F/M/26 Q5 温度降低时内能如何变化 · ■ 43/M/J/22 Q4 烟粒——内能与压强

. A rise in temperature increases the average kinetic energy of the particles, so internal energy rises.

**Temperature** a measure of the average kinetic energy of the particles in a substance

📌 考过 · ■ 42/F/M/26 Q5 粒子平均动能与温度的关系

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**Absolute zero** the lowest possible temperature (−273 °C = 0 K), at which the particles have the least kinetic energy

📌 考过 · ■ 43/O/N/23 Q4 定义绝对零度 + 开尔文↔摄氏度换算

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**Gas pressure** caused by particles colliding with the walls of the container, exerting a force per unit area. Faster particles (higher temperature) or a smaller volume → more frequent, harder collisions → higher pressure.

📌 考过 · ■ 42/F/M/26 Q5 用粒子解释升温时气压变化 · ■ 41/O/N/22 Q4 被困气体——压强与功 · ■ 42/M/J/24 Q4 气压与温度

**Brownian motion** the random motion of (smoke) particles caused by collisions with the (much smaller, fast-moving) molecules of the air or liquid — evidence for the kinetic particle model

📌 考过 · ■ 43/M/J/22 Q4 烟粒随机运动 · ■ 41/M/J/26 Q5 灰尘粒子随机运动

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## FORMULAS

$$T\ (K) = \theta\ (°C) + 273$$

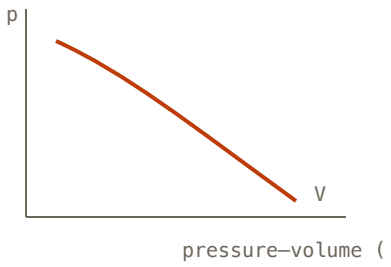
Convert between kelvin and degrees Celsius.

$$pV = \text{constant}$$

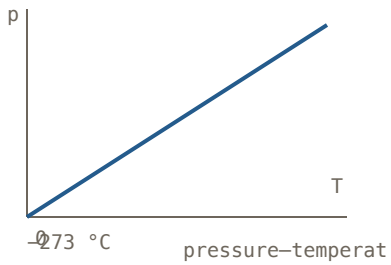
Boyle's law — fixed mass of gas at constant temperature.

📌 考过 · ■ 43/O/N/23 Q4 体积翻倍求新压强 · ■ 42/O/N/25 Q4 气缸内气体 pV 关系 · ■ 42/O/N/22 Q4 粒子动量与压强变化

## GRAPHS




**p–V graph** — a curve falling to the right: pressure is inversely proportional to volume.




**p–T graph** — a straight line through absolute zero: pressure ∝ temperature (fixed volume).

# Thermal Properties & Temperature

2.2

## DEFINITIONS

**Specific heat capacity (c)** the energy required to raise the temperature of unit mass (1 kg) of a substance by 1 °C

📌 考过 · ■ 42/F/M/24 Q4 测油的比热容 · ■ 43/M/J/24 Q4 算加热水所需能量 · ■ 42/O/N/23 Q4 算水升温 20°C 所需能量

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**Melting** solid → liquid at a fixed temperature (0 °C for water); temperature remains constant while energy is absorbed.

📌 考过 · ■ 42/M/J/24 Q4 熔化时温度不变 · ■ 42/O/N/24 Q4 高压锅——沸点与气压

**Boiling** liquid → gas throughout the whole liquid at a fixed temperature (100 °C for water at standard pressure); temperature remains constant while energy is absorbed.

📌 考过 · ■ 42/M/J/24 Q4 沸腾时温度不变 · ■ 42/O/N/24 Q4 高压锅——沸点与气压

**Evaporation** the escape of the more energetic molecules from the surface of a liquid, at any temperature

📌 考过 · ■ 42/F/M/22 Q3 出汗降温 · ■ 41/M/J/22 Q3 分子模型解释蒸发降温 · ■ 42/M/J/26 Q5 蒸发降温与速率

. It cools the remaining liquid (the faster particles leave, so the average kinetic energy falls).

**Condensation / freezing** gas → liquid / liquid → solid; energy is released.

**Thermal expansion** solids, liquids and gases expand when heated — gases expand most, solids least. Used in thermometers and bimetallic strips; expansion gaps are left in bridges and rails.

📌 考过 · ■ 43/O/N/22 Q4 温度计灵敏度与量程 · ■ 41/O/N/24 Q4 热膨胀

## FORMULAS

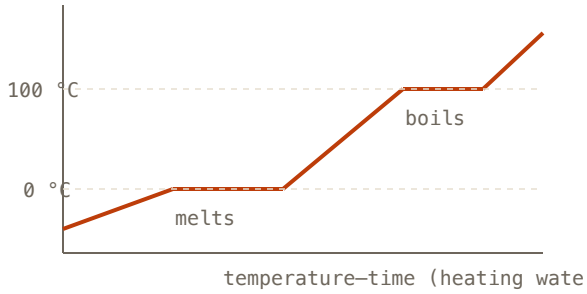
## $\Delta E = mc\Delta\theta$

energy transferred = mass × specific heat capacity × temperature change.

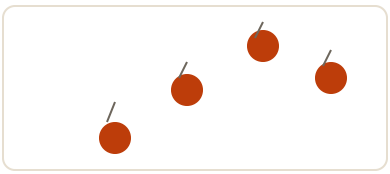
📌 考过 · ■ 42/F/M/24 Q4 E = mcΔθ 计算 · ■ 43/M/J/24 Q4 E = mcΔθ 计算

Rearranged: c = ΔE ⁄ (m Δθ).

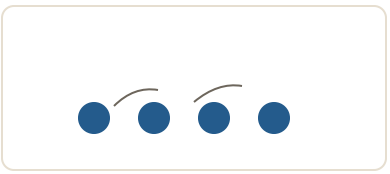
### GRAPH




**Heating curve** — flat regions where melting and boiling occur: energy is absorbed without a temperature rise.



**Evaporation** — faster particles escape the surface; slower ones remain.



**Boiling** — bubbles form throughout the liquid; particles escape everywhere.

# Transfer of Thermal Energy

2.3

## DEFINITIONS

**Conduction** energy transferred through a solid by lattice vibrations (atoms/ions vibrate) and, in metals, by free (delocalised) electrons

📌 考过 · ■ 43/M/J/24 Q5 金属/非金属/气体的传导 · ■ 41/O/N/23 Q2 铜锅——传导 · ■ 43/M/J/25 Q4 铜锅——传导与比热容

. Poor in gases — particles are far apart.

**Convection** energy transferred in a liquid or gas by density changes — hotter (less dense) fluid rises, colder (denser) fluid sinks

📌 考过 · ■ 41/M/J/24 Q3 冰在温水中——对流 · ■ 41/M/J/25 Q4 暖气——对流 · ■ 41/M/J/22 Q6 海陆风——辐射与对流

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**Radiation** infrared (electromagnetic) radiation emitted by all objects; travels through a vacuum — no medium is required

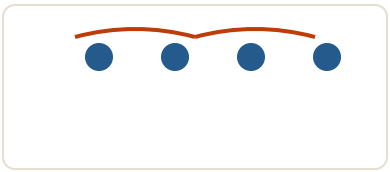
📌 考过 · ■ 42/F/M/26 Q4 户外取暖器——热辐射 · ■ 42/F/M/25 Q3 电暖器——辐射 · ■ 42/M/J/25 Q4 冷却容器——辐射

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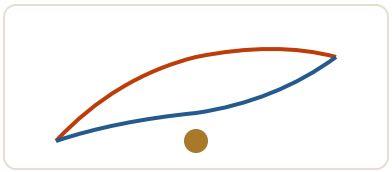
**Emitters & absorbers** dull black surfaces are good emitters and absorbers of infrared; shiny white/silver surfaces are poor emitters and absorbers but good reflectors.

## HOW HEAT MOVES

| Mechanism | Medium | How it works |
|---|---|---|
| Conduction | solids | lattice vibrations + free electrons |
| Convection | liquids, gases | density changes (hot rises, cold sinks) |
| Radiation | vacuum / any | infrared waves; no medium needed |



**Conduction** — energy passes along by particle vibration.



**Convection** — a convection current: hot fluid rises, cool fluid sinks.

## Formula summary

| Quantity | Formula | Notes |
|---|---|---|
| Kelvin ↔ Celsius | $T = \theta + 273$ | absolute zero = 0 K = −273 °C |
| Gas pressure (Boyle's law) | $pV$ = constant | fixed mass, constant temperature |
| Specific heat capacity | $\Delta E = mc\Delta\theta$ | energy to heat mass m by $\Delta\theta$ |

Water: melting point 0 °C, boiling point 100 °C (standard pressure) · specific heat capacity 4200 J/(kg °C).